14.14 Exercise 46: Authentication & Authorization REST API

14.14.1 Description

In this hands-on lab, we will build an **authentication and authorization system** for a REST API using **Laravel 12**, **Sanctum**, and **MySQL**. We use JWT-like tokens for authentication and protect certain API routes.

The API will support:

- **User Registration** (`/api/register`)
- **User Login** (`/api/login`)
- **JWT-based Authentication** with expiration time
- **User Profile Retrieval** (`/api/profile`, protected by JWT token)

We will use **bcrypt** to hash passwords and **Laravel Sanctum** to generate **JWT-like API tokens**.

14.14.2 Objectives

By the end of this lab, you will:

- Activate **API support** in Laravel 12
- Implement **user registration with bcrypt password hashing**
- Implement **JWT authentication using Laravel Sanctum**
- Protect **profile API using JWT token**
- Test the API using **Postman or cURL**

14.14.3 Prerequisites

Before starting, ensure you have:

- **PHP 8.2+**, Composer, and MySQL installed
- **Laravel 12 installed**
- **Postman or cURL** for API testing

> *This lab uses **PHP 8.4**, **Laravel 12** and **Visual Studio Code** as the primary editor.*

14.14.4 Steps